

## 2024 Calculus II    Additional Exercise (7)

1. Exercise 4 in page 38 from Calculus Supplement Homework.
2. (1). Find the unit tangent vector  $\mathbf{T}$  and principal normal vector  $\mathbf{N}$  for the plane curve  $C$ :  $y = x^3$ .  
(2). Find the curvature function  $\kappa(x)$  for the curve  $C$  and find points at which  $\kappa(x)$  has respectively its absolute maximum and minimum values.  
(3). Find the osculating circle for the curve  $C$  at  $x = 1$ .
3. Exercise 24, (1),(3),(5),(7),(9) in page 41 from Calculus Supplement Homework.
4. Exercise 42 in page 43 from Calculus Supplement Homework.
5. Let  $f(x, y) = \frac{\sin xy}{\sqrt{x^2+y^2}}$ . Define the value of  $f(0, 0)$  such that  $f$  is continuous at  $(0, 0)$ .

# Sandwich Thm for Two variables functions

**The Sandwich Theorem** for functions of two variables states that if  $g(x, y) \leq f(x, y) \leq h(x, y)$  for all  $(x, y) \neq (x_0, y_0)$  in a disk centered at  $(x_0, y_0)$  and if  $g$  and  $h$  have the same finite limit  $L$  as  $(x, y) \rightarrow (x_0, y_0)$ , then

$$\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = L.$$

## Q1 P38 Ex4

4. Re-parametrize the curve with respect to the arc length measured starting from  $t = 0$  in the direction of  $t$  increasing:

$$\mathbf{r}(t) = 12t\mathbf{i} + (1 - 3t)\mathbf{j} + (5 + 4t)\mathbf{k}.$$

Sol:

$$\begin{cases} x(t) = 12t \\ y(t) = 1 - 3t \\ z(t) = 5 + 4t \end{cases} \Rightarrow \begin{cases} x'(t) = 12 \\ y'(t) = -3 \\ z'(t) = 4 \end{cases}$$

Arc length parameter:

$$s(t) = \int_0^t \sqrt{[x'(\tau)]^2 + [y'(\tau)]^2 + [z'(\tau)]^2} d\tau$$

$$= \int_0^t 13 d\tau$$

$$= 13t$$

$$\therefore t = \frac{s}{13} \quad \text{代入曲线} \quad \vec{r}(t(s)) = \frac{12}{13}s\vec{i} + (1 - \frac{3}{13}s)\vec{j} + (5 + \frac{4}{13}s)\vec{k}$$

Q2

2. (1). Find the unit tangent vector  $\mathbf{T}$  and principal normal vector  $\mathbf{N}$  for the plane curve  $C: y = x^3$ .
- (2). Find the curvature function  $\kappa(x)$  for the curve  $C$  and find points at which  $\kappa(x)$  has respectively its absolute maximum and minimum values.
- (3). Find the osculating circle for the curve  $C$  at  $x = 1$ .

Sol.: (1)  $\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j}$ , where  $\begin{cases} x(t) = t \\ y(t) = t^3 \end{cases} \Rightarrow \begin{cases} x'(t) = 1 \\ y'(t) = 3t^2 \end{cases}$

$$\frac{d\vec{r}}{dt} = (1, 3t^2)$$

$$\frac{ds}{dt} = \left| \frac{d\vec{r}}{dt} \right| = \sqrt{1 + 9t^4}$$

$$\vec{T} = \frac{d\vec{r}/dt}{|d\vec{r}/dt|} = \frac{1}{\sqrt{1+9t^4}} (1, 3t^2)$$

$$\frac{d\vec{T}}{dt} = \left( \frac{-18t^3}{(1+9t^4)^{\frac{3}{2}}}, \frac{6t\sqrt{1+9t^4} - \frac{3t^2 \cdot 36t^3}{2\sqrt{1+9t^4}}}{(1+9t^4)^{\frac{3}{2}}} \right)$$

$$= \left( \frac{-18t^3}{(1+9t^4)^{\frac{3}{2}}}, \frac{6t + 54t^5 - 3 \cdot 18t^5}{(1+9t^4)^{\frac{3}{2}}} \right)$$

$$= \left( \frac{-18t^3}{(1+9t^4)^{\frac{3}{2}}}, \frac{6t}{(1+9t^4)^{\frac{3}{2}}} \right)$$

$$\left| \frac{d\vec{T}}{dt} \right| = \sqrt{\frac{(6t)^2}{(1+9t^4)^3} ((3t)^2 + 1^2)}$$

$$= \frac{6|t|}{1+9t^4}$$

$$\therefore \vec{N} = \frac{d\vec{T}/dt}{|d\vec{T}/dt|} = \frac{1}{|t|\sqrt{1+9t^4}} (-3t^3, t)$$

(2) by (1) we get

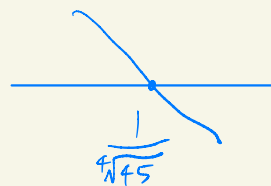
$$K(x) = K(t) = \frac{|d\vec{T}/dt|}{|ds/dt|} = \frac{6|t|}{(1+9t^4)^{\frac{3}{2}}} = \frac{6|x|}{(1+9x^4)^{\frac{3}{2}}}$$

$\therefore K(x)$  is even function, we only need to consider  $x \geq 0$  case.

$$K'(x) = 6 \cdot \frac{(1+9x^4)^{\frac{3}{2}} - \frac{3}{2} \cdot \sqrt{1+9x^4} \cdot 36x^3 \cdot x}{(1+9x^4)^3}$$

$$= 6 \cdot \frac{1+9x^4 - 54x^4}{(1+9x^4)^{\frac{5}{2}}}$$

$$= \frac{6(1-45x^4)}{(1+9x^4)^{\frac{5}{2}}}$$

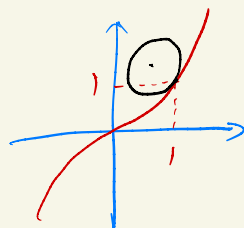


$\therefore$  maximal curvature points:  $(\pm \frac{1}{\sqrt[4]{45}}, \pm \frac{1}{45^{\frac{3}{4}}})$

minimal curvature points:  $(0, 0)$ .

(3)

Osculating circle: 密切圆



$$K(1) = \frac{6}{(1+9)^{\frac{3}{2}}} = \frac{3}{5\sqrt{10}} \Rightarrow \text{radius } \rho(1) = \frac{1}{K(1)} = \frac{5\sqrt{10}}{3}$$

$$\begin{aligned} \text{curvature center: } (1, 1) + \rho(1) \vec{N} &= (1, 1) + \frac{5\sqrt{10}}{3} \cdot \frac{1}{\sqrt{10}} (-3, 1) \\ &= (-4, \frac{8}{3}) \end{aligned}$$

$$\therefore \text{Osculating circle: } (x+4)^2 + (y-\frac{8}{3})^2 = \frac{250}{9}$$

□

Q3 P41 Ex 24 (1) (3) (5) (7)

24. Determine if the following limit exists or not. If so, find the limit. If not, state your reasons:

$$(1) \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^4 + x^2 + y^2};$$

$$(2) \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^{1.5}}{x^4 + y^2};$$

$$(3) \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2};$$

$$(4) \lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x^3 + y^3)}{x^2 + y^2};$$

$$(5) \lim_{(x,y) \rightarrow (0,2)} \frac{\ln(1 + xy^2)}{x};$$

$$(6) \lim_{(x,y) \rightarrow (0,0)} (1 - 2xy)^{1/(x^2 + y^2)};$$

$$(7) \lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x - y)}{x + y};$$

$$(8) \lim_{(x,y) \rightarrow (0,2)} \frac{\arctan \frac{xy}{1+xy}}{x};$$

$$(9) \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{|x| + |y|};$$

$$(10) \lim_{(x,y) \rightarrow (0,0)} \frac{\sqrt{1 + x^2 + y^2} - 1}{x^2 + y^2};$$

Sol: (1)  $0 \leq \frac{x^2 y^2}{x^4 + x^2 + y^2} \leq \frac{x^2 y^2}{\sqrt[3]{x^6 y^2}} = \frac{1}{3} \sqrt[3]{y} \xrightarrow{(x,y) \rightarrow (0,0)} 0$

$\therefore$  原极限为 0

(3) 不存在: along  $x=0 \Rightarrow$  原式  $= 0$

along  $y=x^2 \Rightarrow \lim_{x \rightarrow 0} \frac{x^4}{x^4 + x^4} = \frac{1}{2}$

(5)  $\lim_{(x,y) \rightarrow (0,2)} \frac{\ln(1 + xy^2)}{x y^2} \cdot y^2 = \lim_{(x,y) \rightarrow (0,2)} \frac{\ln(1 + xy^2)}{x y^2} \cdot \lim_{(x,y) \rightarrow (0,2)} y^2 = 4$

(7) 不存在: along  $y=x \Rightarrow$  原式  $= 0$

along  $y=2x \Rightarrow$  原式  $= \lim_{x \rightarrow 0} \frac{\sin(-x)}{3x} = -\frac{1}{3}$

(9)  $0 \leq \frac{x^2 + y^2}{|x| + |y|} \leq \frac{x^2 + 2|x||y| + y^2}{|x| + |y|} \leq |x| + |y|$

$\downarrow$  as  $(x,y) \rightarrow (0,0)$   
0

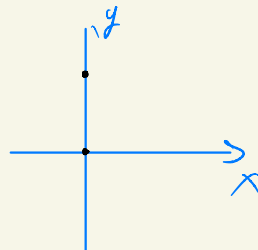
$\downarrow$  as  $(x,y) \rightarrow (0,0)$   
0

# Q4 P43 Ex 42

42. Find all points at which is function is continuous:

$$f(x, y) = \begin{cases} \frac{\sin(xy)}{x}, & x \neq 0 \\ y, & x = 0 \end{cases}$$

Sol: for all  $(x, y)$  with  $x \neq 0$ ,  
 $f(x, y) = \frac{\sin(xy)}{x}$  continuous.  
 Consider point  $(0, y_0)$



$$\textcircled{1} y_0 = 0, \quad 0 \leq \left| \frac{\sin(xy)}{x} \right| \leq |y| \quad \text{near } (0, 0)$$

$\downarrow$   $\downarrow (x,y) \rightarrow (0,0)$   
 $0$   $0$

$$\textcircled{2} y_0 \neq 0, \quad \lim_{(x,y) \rightarrow (0,y_0)} \frac{\sin(xy)}{xy} = 1$$

$$\therefore \lim_{(x,y) \rightarrow (0,y_0)} y \cdot \frac{\sin(xy)}{xy} = y_0$$

Hence  $f(x, y)$  continuous on  $\mathbb{R}^2$ .

□

Q5

5. Let  $f(x, y) = \frac{\sin xy}{\sqrt{x^2+y^2}}$ . Define the value of  $f(0, 0)$  such that  $f$  is continuous at  $(0, 0)$ .

Sol:

$$0 \leq \left| \frac{\sin xy}{\sqrt{x^2+y^2}} \right| \leq \frac{|xy|}{\sqrt{x^2+y^2}} \leq \frac{x^2+y^2}{2\sqrt{x^2+y^2}} = \frac{1}{2} \sqrt{x^2+y^2}$$

$\downarrow (x,y) \rightarrow 0$                        $\downarrow (x,y) \rightarrow (0,0)$

0                                              0

□